# A REMARK ON THE EXPONENT OF THE $O(2^{0.49n})$ SUBSET SUM ALGORITHM: 0.4889, AND 0.4741 FOR DENSE INSTANCES

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ABSTRACT. The OpenAI Math Release preprint *Subset Sum in Time* $O(2^{0.49n})$ gives a randomized worst-case algorithm for Subset Sum with running time $O(2^{0.49n})$. **Assuming that preprint is correct**, we show that its method gives $O(2^{0.4889n})$ in the same model. The changes are: re-optimized parameters, a one-parameter variant of the alias-extraction step (the shared block is split in proportion $\sigma = 0.38$ instead of $1/2$, at the cost of exponentially many trials), and new eighth-order filtering densities with two different tilting exponents. The new filtering moments are certified rigorously with interval arithmetic. For instances with $\max\{t, a_i\} \le 2^{n/2}$ the checksum alone gives $O(2^{0.4741n})$. This is a re-tuning of the original method, not a new method; a heuristic limit of the same framework is about 0.4798. We also record an elementary fact about finite abelian $p$-groups (Proposition 1.3): homomorphic reductions of $(\mathbb{Z}/p^e)^k$, $e \ge 2$, cannot replace the random-prime collision bound used in the preprint.

**Attribution and premise.** The algorithm, its analysis and all lemmas we cite are OpenAI's work: family 138 of `https://github.com/openai/math` (commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`), preprint [1]. *We take the correctness of* [1] *as given*; it has no formal proof and, to our knowledge, no published peer review. We refer to its statements by number (Theorem 1.1, Proposition 2.4, Lemma 2.5, Proposition 3.1, Lemma 3.2, Theorem 4.1, Lemma 4.2, Theorem 5.1, Lemma 5.2). This note only changes parameters and one sampling rule, and redoes the bookkeeping. Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; independent adversarial reviews were run by separate AI sessions and are not human peer review.

## 1. STATEMENTS

The model is exactly that of [1, Section 1.1]: a word RAM with word length $w = \lceil 4(n + b + \log_2(n+2)) \rceil$, where $b$ is the maximum bit length of the input.

**Theorem 1.1.** *Assume Theorem 1.1 of* [1] *and its proof. There is a uniform randomized algorithm for Subset Sum that halts on every execution, answers correctly with probability at least* $2/3$*, and for every fixed* $c$ *runs on every execution in at most* $C_c 2^{0.4889n}$ *word operations on inputs with* $b \le n^c$ *and* $n \ge N_c$*.*

**Corollary 1.2** (dense instances)**.** *Under the same premise, if moreover* $\max\{t, a_1, \dots, a_n\} \le 2^{n/2}$*, the time bound improves to* $O(2^{0.4741n})$*. With the filtering densities of* [1] *the same argument gives about* $2^{0.4748n}$ *(numerical value* 0.47474*);* [1] *does not state a dense-case bound.*

**Proposition 1.3.** *Let* $G = (\mathbb{Z}/p^e)^k$ *with* $p$ *prime,* $e \ge 1$*, and let* $K \le G$ *have index* $M$*. Write* $G[p] = \{g : pg = 0\}$*. Then*

$$|K \cap G[p]| \;\ge\; p^k \, M^{-1/e},$$

*with equality if and only if* $G/K \cong (\mathbb{Z}/p^e)^s$ *for some* $s$*. Consequently, for every probability distribution on subgroups* $K$ *of index at most* $M$ *(equivalently, on homomorphisms* $\psi$ *from* $G$ *onto groups*

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*of order at most $M$, with $K = \ker\psi$) there is $0 \neq g \in G[p]$ with*

$$\Pr[g \in K] \;\geq\; \frac{p^k M^{-1/e} - 1}{p^k - 1}.$$

For $e \geq 2$ and $M \leq p^k$ the right side is of order $M^{-1/e}$, far above the $\approx M^{-1}$ that a random prime modulus achieves for a fixed nonzero integer in [1, Lemma 2.5]. So the step of [1, Section 5] that bounds visited pairs through Lemma 2.5 has no direct analogue for subset sum over $(\mathbb{Z}/p^e)^k$ when the modulus is replaced by a homomorphic reduction. This is an obstruction to one proof step, not a lower bound for any problem.

## 2. Alias extraction with an unbalanced shared block

Fix

$$\eta = 0.5176, \quad \sigma = \tfrac{19}{50}, \quad \mu = 0.0877, \quad m = 50\lfloor \mu n/50 \rfloor,$$
$$\gamma = 0.3714, \quad \delta = 0.0112, \quad \theta' = 0.0226, \quad K = 15,$$

so that $\sigma m \in \mathbb{Z}$. Write $h$ for the binary entropy and $\rho = (1 - h(\sigma))\mu$. Everything not mentioned below is as in [1, Section 5] (with .0567, .48925, .51772, 17 replaced by $\mu, \gamma, \eta, K$).

*Trials.* A trial draws $M, F$ as in [1]. The representations use the event $|J \cap M| = \sigma m$ instead of $|J \cap M| = m/2$. In layer $k \in \{0, \dots, \sigma m\}$ left masks have size $k$ and right masks size $\sigma m - k$, so $N_L = 2^l \binom{m}{k}$, $N_R = 2^r \binom{m}{\sigma m - k}$, balanced as in [1, (5.6)]. For every target $s$ and carry $q$ the procedure runs

$$T = n^{10} \left\lceil 2^m / \tbinom{m}{\sigma m} \right\rceil = 2^{\rho n + O(\log n)}$$

independent trials (instead of $n^{10}$). This number is a fixed function of $n$.

*Cumulative caps (repair A).* In [1] each filtered list and each visited-pair count is capped at $2^{\lceil \tau n \rceil}$; with polynomially many trials this bounds the total work. With $T$ exponential we instead impose one budget $2^{\lceil \tau_0 n \rceil}$, $\tau_0 = 0.48888$, on the *total* number of emitted filtered records and one on the *total* number of visited pairs, summed over all $s, q$, trials and layers. The procedure fails as soon as a budget would be exceeded.

**Lemma 2.1** (sizes)**.** *Without any assumption on the input other than the guard $b \leq 2^{10^{-5} n}$ of* [1, (5.1)]*,*

$$\log_2 \mathbb{E}[\textit{total filtered records}] \leq \big(\rho + \tfrac{1-\mu}{2} + \mu h(\tfrac{\sigma}{2}) - \gamma\mu\big)n + O(\log n) \;<\; 0.488777\, n,$$
$$\log_2 \mathbb{E}[\textit{total visited pairs}] \leq \big(\rho + 1 - \mu + 2\mu h(\tfrac{\sigma}{2}) - \gamma\mu - \eta + 10^{-5}\big)n + O(\log n) \;<\; 0.488857\, n,$$

*and list generation costs $2^{(\rho+\mu+1/4)n + O(\log n)}\,\mathrm{poly}(b) \leq 2^{(\rho+\mu+1/4+10^{-5})n + O(\log n)} < 2^{0.3414 n}$ on every execution.*

*Proof.* For one trial and one layer the computation is that of [1, (5.8), (5.11)], with $\binom{m}{k}\binom{m}{\sigma m - k} \leq 2^{m(h(k/m) + h(\sigma - k/m))} \leq 2^{2mh(\sigma/2)}$ by concavity of $h$. In particular a filtered list has expected length at most $2\sqrt{N_L N_R}/P' \leq 2^{(n-m)/2 + mh(\sigma/2) - \gamma m + O(1)}$, and a fixed raw pair is visited with probability at most $\mathrm{poly}(n)(b + \log n)2^{-\eta n} \cdot 2^{1-\gamma m}$ by [1, Lemma 2.5] and the uniform residue $j'$; the factor $b + \log n \leq 2^{10^{-5} n}\,\mathrm{poly}(n)$ is the guard term, which the visited-pair exponent must carry (repair B). Multiply by $T$ and by the polynomial number of $(s, q, k)$. Generation per trial is $2^{m + \lceil \max(l,r)/2 \rceil}\,\mathrm{poly}(n+b)$ as in [1, (5.7)]. The numerical values are certified in `chain_check.py`. □

By Markov's inequality each budget is exceeded with probability $2^{(0.488857 - 0.48888)n + O(\log n)} = o(1)$. Coupling the capped and uncapped procedures as in [1, end of Section 5] transfers the coverage statement below.

**Lemma 2.2** (coverage)**.** *Assume* $\mathbb{E}_{|V|=\lfloor n/2\rfloor} d(V) \le \delta n$ *and the event of* [1, Lemma 5.2] *(which here holds with probability* $1-o(1)$ *because* $h(\mu)+\mu+10^{-5} = 0.516456 < \eta$*). Then, with probability* $1-o(1)$*, the uncapped procedure recovers every alias of every target.*

*Proof.* Fix an alias $I$ and its carry. In one trial $J = I \triangle F$ is uniform and independent of $M$, so $\Pr[|J\cap M| = \sigma m] = \binom{m}{\sigma m} 2^{-m}$ and, conditionally, $S = J \cap M$ is a uniform $\sigma m$-subset of $[n]$. Since $K\sigma\mu = 0.49989 < 1/2$, a uniform $\lfloor n/2\rfloor$-set contains $K = 15$ disjoint sets with the law of $S$ for large $n$; superadditivity of deficiency [1, Lemma 2.2] gives $\mathbb{E}d(S) \le \delta n/15$. With $\theta = \delta/(15\sigma\mu) = 0.022405$, Markov's inequality gives $\Pr[d(S) \le \theta'\sigma m] \ge 1 - \theta/\theta' - o(1) \ge 0.0086 - o(1)$. (Taking $\theta' = \theta$ would make this bound vacuous; a margin $\theta' > \theta$ is needed.) On this event the signed reduced weights of subsets of $S$ take at least $2^{\sigma(1-\theta')m}$ values, so some layer has $B \ge 2^{\sigma(1-\theta')m}/(\sigma m+1) \ge 2^{\gamma m}$ distinct weights, because $\gamma < \sigma(1-\theta') = 0.371412$. From here the argument of [1, (5.17) and the Cauchy–Schwarz step after it] is unchanged: with probability at least $1/(8K_0n^2)$ over $(P', j')$ some representation survives the smaller modulus and is visited. Hence one trial recovers $I$ with probability at least $p_1 = \binom{m}{\sigma m}2^{-m}\cdot 0.008/(8K_0n^2)$, and $T$ independent trials miss it with probability at most $e^{-p_1T} \le e^{-n^7}$. A union bound over the $(nN_0+1)2^n$ pairs (target, subset) finishes the proof. □

The preliminary search [1, Proposition 2.4] is used with 0.01004 replaced by $\delta$: the Chebyshev threshold becomes $\delta - 1.5\cdot 10^{-5}$, and when it returns CONTINUE with non-negligible probability the hypothesis of Lemma 2.2 holds. As in the paper, the cap must also absorb the $+\frac12$ in $\log_2|\Sigma(U)| \le (\frac12 - \delta + 1.5\cdot 10^{-5})n + \frac12$, so we cap at $2^{\lceil 0.488825n\rceil}$.

The prime $P'$ is now drawn $2^{\rho n}\,\mathrm{poly}(n)$ times (once per layer, trial, target and carry). With the bounded sampler of [1, Section 6] ($n^3$ candidates, trial division up to $2^{(\gamma m+1)/2}$) this costs $2^{(\rho+\gamma\mu/2)n+O(\log n)} = 2^{0.01997n+O(\log n)}$ in total, and the failure term of the coupling [1, (6.4)] becomes $2^{\rho n}\,\mathrm{poly}(n)e^{-c_0n^2} = o(1)$.

## 3. New filtering densities and the checksum

In [1, Section 3] replace $r, s$ by the degree-8 cosine polynomials with coefficients

$$(r_j) = (-1.0691,\ -0.569,\ 0.4495,\ 0.4334,\ 0.3313,\ -0.1504,\ -0.1634,\ 0.0317),$$
$$(s_j) = (-0.882,\ 0.6245,\ 0.4375,\ -0.2822,\ 0.2331,\ 0.2563,\ -0.1326,\ -0.1861),$$

and the single tilt $\kappa = 1.0315$ by $f_* = f^{\kappa_f}/\int f^{\kappa_f}$, $g_* = g^{\kappa_g}/\int g^{\kappa_g}$ with $\kappa_f = 1.034$, $\kappa_g = 1.0339$. Nothing in [1, Sections 3–4] uses $\kappa_f = \kappa_g$: $D$, $D-A$, $D-E$ are still relative entropies, and the score identities of [1, Appendix A] become

$$u = L + (\kappa_f - 1)B_f + (\kappa_g - 1)B_g - H, \qquad v = (2-\kappa_f)B_f + (2-\kappa_g)B_g - 2L + H,$$

with $B_f, B_g$ the $K_x$-means of $\log_2 f(x+\cdot)$, $\log_2 g$, and $H = H_f + H_g$. Run Theorem 4.1 with $(\eta,\tau) = (0.5176, \tau_0)$, $\tau_0 = 0.48888$ (repair C: the checksum must be certified at the exponent at which it is run, not at the headline 0.4889), and with the constants 8.5, 8.3 of [1, (3.6), (4.22)–(4.24)] replaced by $\lambda_1 = 8.323$, $\lambda_2 = 8.132$.

**Proposition 3.1.** *With these data,* $\Delta = \eta - \tau_0$*,* $a = \Delta/(2\tau_0 - \eta)$ *and* $M(x)^2 = 2 + 2\cos x$*,*

$$\log_2 \int_{\mathbb{T}} M^2\, 2^{\lambda_1(\Delta - u)} < 0.5170788, \qquad \log_2 \int_{\mathbb{T}} M^2\, 2^{\lambda_2(av - u)} < 0.5169604.$$

*Certificate.* As in [1, Appendix A]: trapezoid sums in interval arithmetic, the analytic trapezoid bound of Trefethen–Weideman [2] for the inner integrals (strip 0.5), and an analytic strip of half-width $2\beta = 0.38$ for the outer integrands, using $|Q(z)-1| \le \|f(\cdot + iy/2) - 1\|_2 \|g(\cdot - iy/2) - 1\|_2 < 0.5654$. Script `hardcore_certificate.py` (mpmath intervals, $N = 128$); an independent implementation (python-flint balls, $N = 192$, strips 0.4 and 0.34, scores computed from the definitions

of $D, A, E$, its own strip bounds) gives the same bounds to 12 digits. This agreement is expected, since both evaluate the same integrals with quadrature errors below $10^{-14}$; the independence is in the code, the arithmetic library and the error analysis. □

Lemma 3.2 of [1] needs only bounded coefficients ($\sum|r_j| = 3.1978$, $\sum|s_j| = 3.0343$), so the finite-circle moments are below $2^{0.51718}$ for $P \geq P_0$. The two domination steps [1, (4.22)–(4.23)] need $1 < \lambda_1 \leq 10$ and $\lambda_2(4a+3) < 30$; here $\lambda_2(4a+3) = 26.43$. The error exponent of [1, (4.25)] becomes $-\eta + 0.51718 + 48\epsilon < -0.00042$, and the checksum runs in $2^{(\tau_0+20\epsilon)n+O(\sqrt{n})}$.

*Proof of Theorem 1.1.* Assemble as in [1, Section 6]. On every guarded execution the exponents of the components are 0.488825 (preliminary cap), 0.48888 (cumulative alias budgets), 0.3414 (generation), 0.01997 (sampling the primes $P'$), $\tau_0 + 20\epsilon = 0.4888802$ (checksum) and $\eta/2$ (trial division for $P$); the largest is $0.4888802 < 0.4889$, and the gap $1.98 \cdot 10^{-5}$ absorbs $\mathrm{poly}(n+b+2)$ when $b \leq n^c$. Correctness is that of [1], with Lemmas 2.1–2.2 in place of Theorem 5.1 and Proposition 3.1 in place of Proposition 3.1. □

*Proof of Corollary 1.2.* Take $\eta_d = 0.50001$ and $P = 2^{\lceil \eta_d n \rceil}$; Theorem 4.1 needs no primality. After the isolation step of [1, Lemma 2.1] every transformed subset sum and target lies in $[0, 2^{n/2+O(\log n)})$, which is contained in $[0, P)$ for large $n$, so the modular bins are exact and no aliases exist. Run Theorem 4.1 for the $O(n^2)$ targets with $(\eta_d, \tau) = (0.50001, 0.47409)$ and $(\lambda_1, \lambda_2) = (8.447, 8.236)$ (the checksum must run strictly below the claimed exponent, since its time is $2^{(\tau+20\epsilon)n+O(\sqrt{n})}$); the certified moments are $< 2^{0.4997104}$, the error exponent is $-\eta_d + 0.4998104 + 48\epsilon < -0.00019$, $\lambda_2(4a+3) = 26.61 < 30$, and the threshold test of [1, Section 6] decides. The time is $2^{0.4740902n+O(\sqrt{n})}\,\mathrm{poly}(n+b) = O(2^{0.4741n})$. □

## 4. Proof of Proposition 1.3

Since $G$ is homocyclic, $G[p] = p^{e-1}G$ and $|G[p]| = p^k$. Let $\pi : G \to G/K$. Then $\pi(G[p]) = p^{e-1}(G/K)$. Write $G/K \cong \bigoplus_j \mathbb{Z}/p^{a_j}$ with $1 \leq a_j \leq e$ and let $s = \#\{j : a_j = e\}$. Then $p^{e-1}(G/K)$ has order $p^s$, and $M = |G/K| \geq p^{es}$, with equality iff all $a_j = e$. Hence $|K \cap G[p]| = |\ker \pi|_{G[p]}| = p^k/p^s \geq p^k M^{-1/e}$, with the stated equality case. For the second claim, $\sum_{0 \neq g \in G[p]} \Pr[g \in K] = \mathbb{E}|K \cap G[p]| - 1 \geq p^k M^{-1/e} - 1$ (the bound is decreasing in $M$, so index $\leq M$ suffices), and the maximum is at least the average. □

The script `theorem_c_exhaustive.py` checks the inequality, the equality cases and the averaged bound over all subgroups of $(\mathbb{Z}/p^e)^k$ for $(p^e, k) \in \{(4,2), (4,3), (8,2), (16,2), (9,2), (27,2), (25,2)\}$; the independent script enumerates the same subgroups as kernels of all endomorphisms. The fact is elementary and we do not claim it is new.

## 5. Where the gain comes from, and the limit of the framework

At the paper's operating point, the alias side and the checksum are both tight ([1, (5.8), (5.11)]: 0.489909, 0.489849; checksum 0.48996). Numerically (not certified), re-optimizing $(\lambda, \eta, \mu, \delta)$ with the paper's densities gives 0.48977; the $\sigma$-variant brings the alias side to 0.48911; the new densities move the checksum curve $\tau_{cs}(\eta)$ down by about $7 \cdot 10^{-4}$ (at $\eta = 0.5176$: $0.48955 \to 0.48882$), giving 0.48881, certified here with margin at $\tau_0 = 0.48888$. In terms of the saving below the meet-in-the-middle exponent 1/2: $0.0100 \to 0.0111$; the running-time ratio is $2^{0.0011n}$.

These are parameter choices and one small variant inside the same proof; no new idea is involved. A heuristic calculation (CONST-OPT notes, not part of this remark) puts the limit of this framework near 0.4798: as $\delta \to 0$ the alias side tends to 0.48803, and an idealized checksum meets the alias-count lower bound $1 - \eta$ near $\eta \approx 0.520$. The paper uses 49.5% of the heuristic headroom $1/2 - 0.4798$, this note 55%. Going substantially below 0.4889 would need a new ingredient (more selective filters or a less wasteful representation), not further tuning.

*Not claimed.* No explicit $N_c$: with $\epsilon = 10^{-8}$ the bounds take effect only for astronomically large $n$ (already in [1]). Nothing is formalized. Related work: Ye and Li [3] independently obtain $O(2^{0.499999n})$ by a different method; Chen, Jin, Randolph and Servedio [4] save a polynomial factor.

## References

[1] OpenAI, *Subset Sum in Time* $O(2^{0.49n})$, OpenAI Math Release preprint, October 4, 2026, https://github.com/openai/math/tree/main/preprints/Subset-Sum-in-Time-2-power-0-49n-October-4-2026.
[2] L. N. Trefethen and J. A. C. Weideman, The exponentially convergent trapezoidal rule, *SIAM Review* 56 (2014), 385–458.
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[4] X. Chen, Y. Jin, T. Randolph and R. A. Servedio, Subset sum in time $2^{n/2}/\mathrm{poly}(n)$, arXiv:2301.07134 (2023).